Arun Dinesh Jana

Software Engineer | C++ | Python | Automotive Software

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LinkedIn Profile



Summary

Software Engineer with one year of professional experience developing C++ and Python software for autonomous-driving systems at Aptiv. Experienced in obstacle-detection software, ROSbag data processing, sensor-data analysis and software quality checks. Currently completing an M.Sc. in Computer Simulation in Science at Bergische Universität Wuppertal, with expected graduation in February 2027. Seeking full-time software engineering roles involving C++, automotive systems or performance-oriented applications.

Technical Skills

Programming: C, C++, C++17, Python, MATLAB
Build and CI Tools: CMake, Jenkins, Git
Testing and Quality: Unit testing (university/personal projects), software quality assurance, automated checks
Containers and Orchestration: Docker, Kubernetes (university/personal projects)
Parallel Computing: MPI, parallel algorithms, performance benchmarking
Operating Systems: Linux, Windows, macOS
Additional Tools and Technologies: Jira, cloud data platforms and APIs, CUDA (introductory)

Projects and Research

Master's Thesis – Non-Dominance Filtering in Parallel June 2026 – Present

- Implemented MAXIMA2 and its recursive FILTER merge step, including a linear-time 2D sweep base case, in C++17.
- Implemented and validated a brute-force $O(N^2)$ baseline using hand-built correctness test cases.
- Developing MPI / CUDA-parallel versions of both algorithms: divide-and-conquer recursion distributed as a binary process tree, and brute-force domination checks distributed through data-parallel processing.
- Benchmarked runtime scaling across point count, dimension and process count to distinguish algorithmic efficiency from parallelization effects.

FHP-1 Lattice Gas Simulation Feb 2025
M.Sc. Project – Computer Simulation in Science, Bergische Universität Wuppertal

- Developed a lattice-gas cellular automaton simulation in C to model fluid dynamics on a two-dimensional hexagonal grid using FHP-1 collision and streaming rules.
- Implemented MPI-based domain decomposition and inter-process boundary exchange to support parallel simulation across multiple processes.
- Modelled flow boundary conditions and obstacles, and analysed macroscopic flow behaviour through spatial averaging and MATLAB visualization.
- Validated simulation correctness and evaluated parallel performance.

Technologies: C, MPI, MATLAB, numerical simulation, cellular automata, parallel computing

Automatic Vehicle Accident Detection and Alert System March 2022
R.M.K College Of Engineering and Technology

- Designed an Arduino-based system using accelerometer, gyroscope, GPS, GSM and proximity sensors to detect potential vehicle accidents.
- Incorporated automatic emergency alerts with GPS location, triggered when the alert is not cancelled within a predefined time.
- Designed to notify emergency contacts and responders to support faster medical assistance during the Golden Hour.

Technologies: Arduino, Embedded Systems, GPS, GSM, Sensors

Professional Experience

Aptiv Wuppertal, Germany
Software Engineer – AS&UX Team, Autonomous Driving Software May 2023 – April 2024

- Developed C++ and Python software for over-drivable and under-drivable obstacle detection in autonomous-driving systems.
- Built a Python tool to convert ROSbag data into a more compatible format, reducing compatibility issues by 80%.
- Analysed ROSbag sensor data in MATLAB, improving data-processing efficiency by 30%.
- Performed quality checks on autonomous-driving algorithms against industry-standard programming rules, improving maintainability by 25%.
- Supported AI-module training through data preprocessing and model refinement, improving object-recognition accuracy by 15%.
- Uploaded and managed large datasets on a cloud platform through APIs to improve data accessibility for autonomous-driving pipelines.
- Used existing Jenkins pipelines and reviewed build or check results as part of the software development workflow.

Bergische Universität Wuppertal Wuppertal, Germany
IT Support and System Administrator (Student / Research Assistant) August 2026 – Present

- Provide first- and second-level IT support for hardware, software and network-related issues.
- Install, configure and maintain Windows and Linux workstations and university IT systems.
- Manage user accounts and access permissions; support software deployment, system maintenance and technical documentation.

CPW GmbH Wuppertal, Germany
Working Student – Intellectual Property Operations November 2024 – April 2025

- Monitored online marketplaces for potential unauthorized use of protected products, brands and intellectual property.
- Investigated and documented potential infringements, digitized operational records and maintained databases to support accurate case management.

Andhra Pradesh Power Generation Corporation Limited
Student Intern September 2021 – October 2021

- Built a Python-based user interface for thermodynamic calculations, reducing manual computation time by 70%.
- Ran MATLAB simulations for turbine and generator performance, improving energy-output predictions by 10%.

Education

Bergische Universität Wuppertal October 2022 – February 2027
M.Sc. Computer Simulation in Science
Expected graduation: February 2027

R M K College of Engineering and Technology 2018 – 2022
B.E. Mechanical Engineering

Publications and Awards

Publications

- "Capsulization of Vertical Axis Wind Turbine"
- "IoT-Based Vehicle Accident Detection and Alerting System"

Awards

- Best Paper Presentation – Industry 4.0
- First Prize – Solving Mechanical Problems Using Programming Year: 2020
- First Prize – Solving Mechanical Problems Using Programming Year: 2019

Languages

- English (Full Professional Proficiency)
- German (B1 / Working Proficiency)
- Telugu and Tamil (Native or Bilingual Proficiency)